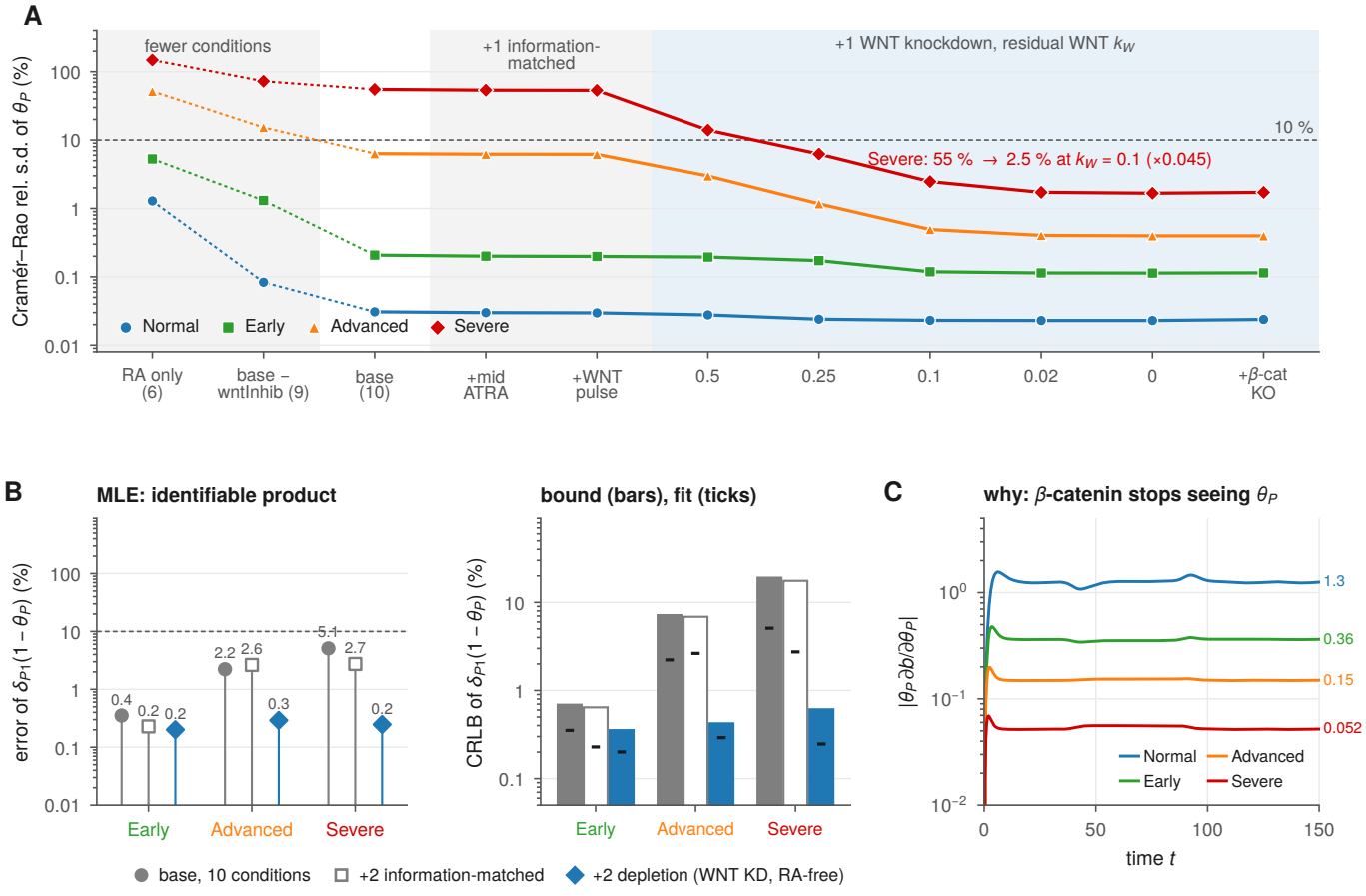
A
fewer conditions
+1 information-matched
+1 WNT knockdown, residual WNT k_W
Cramér–Rao rel. s.d. of θ_P (%)
100
10
1
0.1
0.01
10 %
Severe: 55 % → 2.5 % at k_W = 0.1 (×0.045)
Normal
Early
Advanced
Severe
RA only (6)
base – wntInhib (9)
base (10)
+mid ATRA
+WNT pulse
0.5
0.25
0.1
0.02
0
+β-cat KO
B
MLE: identifiable product
error of $\delta_{P1}(1-\theta_P)$ (%)
0.4
0.2
0.2
2.2
2.6
0.3
5.1
2.7
0.2
Early
Advanced
Severe
bound (bars), fit (ticks)
CRLB of $\delta_{P1}(1-\theta_P)$ (%)
base, 10 conditions
+2 information-matched
+2 depletion (WNT KD, RA-free)
C
why: β-catenin stops seeing θ_P
$|\theta_P \partial b/\partial\theta_P|$
10^0
10^{-1}
10^{-2}
1.3
0.36
0.15
0.052
Normal
Advanced
Early
Severe
0
50
100
150
time t